

41.3 Likelihood under missingness at random

A complete-case analysis throws away recorded values: a unit whose income is missing still told us its age. Likelihood throws nothing away. It asks what the *observed* data were likely to be, and the observed data include the partial records. This section writes that likelihood down, proves the theorem that makes it usable, and says what the theorem costs.

41.3.1 The observed-data likelihood

The full model is Eq. 41.1.1: a density for the complete array and a mechanism. Since $\mathbf{D}_{\text{mis}}$ was never seen, the likelihood of what was seen integrates it out,

$$L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) = \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}}, \boldsymbol{\psi}) d\mathbf{D}_{\text{mis}}, \quad (41.3.1)$$

a function of both parameters. Beside it stands the much simpler

$$L(\boldsymbol{\theta}) = f(\mathbf{D}_{\text{obs}} \mid \boldsymbol{\theta}) = \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) d\mathbf{D}_{\text{mis}}, \quad (41.3.2)$$

which ignores the mechanism entirely. The **observed-data likelihood** Eq. 41.3.2 is what software computes when told that some entries are absent. The question is when using it is the same as using Eq. 41.3.1.

Theorem 41.3.1. Ignorability

Suppose that

- (i) the mechanism is MAR at the realized $(\mathbf{R}, \mathbf{D}_{\text{obs}})$ in the sense of Definition 41.1.1(b);
- (ii) the parameters are **distinct**: the parameter space of $(\boldsymbol{\theta}, \boldsymbol{\psi})$ is a product set $\Theta \times \Psi$, so that knowing one places no restriction on the other.

Then the full likelihood factors,

$$L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) = L(\boldsymbol{\theta}) \cdot f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}), \quad (41.3.3)$$

and consequently:

- (a) the maximizer of L_{full} in $\boldsymbol{\theta}$ is the maximizer of L , and likelihood ratio statistics for hypotheses about $\boldsymbol{\theta}$ alone are the same under both;
- (b) if the prior is $p(\boldsymbol{\theta}, \boldsymbol{\psi}) = p(\boldsymbol{\theta})p(\boldsymbol{\psi})$, the posterior of $\boldsymbol{\theta}$ given $(\mathbf{D}_{\text{obs}}, \mathbf{R})$ equals its posterior given $\mathbf{D}_{\text{obs}}$ alone, computed from Eq. 41.3.2;
- (c) if in addition the MAR condition holds at every $(\mathbf{R}, \mathbf{D}_{\text{obs}})$, the observed-data score has mean zero and the usual maximum likelihood asymptotics (Theorem 7.3.1) apply to L under the ordinary regularity conditions.

A mechanism satisfying (i) and (ii) is called **ignorable**.

Proof. Under (i) the factor $f(\mathbf{R} \mid \mathbf{D}, \boldsymbol{\psi})$ in Eq. 41.3.1 takes the same value for every $\mathbf{D}_{\text{mis}}$, namely $f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$, so it comes out of the integral:

$$\begin{aligned} L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) &= f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}) \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) d\mathbf{D}_{\text{mis}} \\ &= f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}) L(\boldsymbol{\theta}), \end{aligned}$$

which is Eq. 41.3.3.

- (a) By (ii), $\boldsymbol{\theta}$ ranges over Θ whatever $\boldsymbol{\psi}$ is, so $\max_{\boldsymbol{\psi}} L_{\text{full}} = L(\boldsymbol{\theta}) \max_{\boldsymbol{\psi}} f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$, and the second factor does not involve $\boldsymbol{\theta}$. Maximizing over $\boldsymbol{\theta}$ therefore maximizes L . For a hypothesis $\boldsymbol{\theta} \in \Theta_0$, both the numerator and the denominator of the likelihood ratio carry the same second factor, which cancels.
- (b) The posterior is proportional to $p(\boldsymbol{\theta})p(\boldsymbol{\psi})L_{\text{full}}$. By Eq. 41.3.3 this is $\{p(\boldsymbol{\theta})L(\boldsymbol{\theta})\} \cdot \{p(\boldsymbol{\psi})f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})\}$, a product of a function of $\boldsymbol{\theta}$ and a function of $\boldsymbol{\psi}$. Integrating out $\boldsymbol{\psi}$ multiplies the first bracket by a constant, so the marginal posterior of $\boldsymbol{\theta}$ is proportional to $p(\boldsymbol{\theta})L(\boldsymbol{\theta})$.
- (c) If (i) holds at every realization, then at every realization the joint density of the observed data $(\mathbf{D}_{\text{obs}}, \mathbf{R})$ is exactly $L(\boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$, whose logarithm is $\log L(\boldsymbol{\theta})$ plus a term free of $\boldsymbol{\theta}$. So $\partial \log L / \partial \boldsymbol{\theta}$ is the $\boldsymbol{\theta}$ component of the score of a correctly specified density for the data actually seen, and has mean zero under the ordinary regularity conditions of Theorem 7.3.1.

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Remark (Where each hypothesis is used). Condition (i) alone gives Eq. 41.3.3 and part (b): a Bayesian who conditions on the data in hand needs nothing more. Condition (ii) is what makes the factorization useful to a frequentist, who maximizes; without it the two factors share parameters and neither can be maximized alone. Part (c) needs the everywhere version of MAR, because a sampling distribution averages over data sets that did not occur. Textbooks usually state “MAR plus distinctness” without separating these roles; the separation is due to Seaman, Galati, Jackson and Carlin (2013).

Idea. Under MAR with distinct parameters, the analyst may forget the mechanism and maximize the likelihood of what was recorded. That is the entire licence for the methods of the next two sections — and it is a licence to use a likelihood that is usually an integral, not a product.

41.3.2 What the licence costs

Ignorability licenses ignoring the mechanism, not ignoring the missing values. Three consequences follow.

A model is now needed for variables previously conditioned on. In a complete data set, least squares never models the distribution of $\mathbf{X}$: the likelihood conditions on it. When a covariate is missing, Eq. 41.3.2 integrates over it, and that integral needs a density. Missing responses cost nothing extra; missing covariates cost a model for the covariates.

When only the response is missing, the partial records add nothing. If every covariate is recorded, the integral in Eq. 41.3.2 over a missing y_i is one, and the observed-data likelihood is exactly the complete-case likelihood. This is Proposition 41.2.1(b) again, as a statement about likelihoods, and it is the situation of Theorem 18.6.1.

The likelihood is no longer a product of simple factors. Each incomplete record contributes a marginal density, obtained by integrating the complete-data density over the entries it lacks. For the multivariate normal those integrals are marginals of a normal; for almost anything else they have no closed form. Section 41.4 is the standard response.

For a **monotone** pattern the integrals are free: integrating out a record's unrecorded tail deletes exactly its conditional factors, so the likelihood becomes a chain of conditional models on nested subsamples, each maximized on its own; that is Exercise (B1) of Section 41.1. The bivariate case is the subject in miniature. With u always recorded and v recorded on the first k units, the estimate of μ_v is $\bar{v}_{(k)} + \hat{\beta}\{\bar{u}_{(n)} - \bar{u}_{(k)}\}$ by Exercise (B1) below: the complete-case mean of v , corrected by the shift in the mean of u , using the $n - k$ records that carry a u and no v . Those records were useless to the complete-case analysis and are decisive here — and they repair the damage only to the extent that u predicts v , a remark that returns, with teeth, in Section 41.5. Anderson (1957) solved the general monotone multivariate normal case this way.

41.3.3 What changes under MNAR

Drop the MAR assumption and Eq. 41.3.3 fails: the integral in Eq. 41.3.1 cannot release the mechanism, and the two parameters are entangled. The mechanism must then be modelled, and the model is not checkable, since Proposition 41.1.1 says that varying the assumed conditional distribution of the missing values traces out a family with identical observed-data likelihoods. Whatever identification such a model achieves comes from functional-form assumptions rather than from the data, and the likelihood surface is often nearly flat in the directions that matter. Section 41.6 takes this up as a problem of reporting rather than of estimation.

41.3.4 Exercises

41.3.4.1 A. Check your understanding

Exercise. A1

Give a mechanism under which MAR holds but the parameters are not distinct, and show that Eq. 41.3.3 still holds while part (a) of Theorem 41.3.1 can fail. (Hint: let the probability of recording y_i be a known function of β .)

Solution. Take $\Pr(R_i = 1 \mid \mathbf{x}_{(i)}) = \Phi(\mathbf{x}_{(i)}^\top \beta)$, with no free ψ at all. The mechanism depends only on recorded covariates, so MAR holds and the factorization Eq. 41.3.3 is valid, but the second factor now depends on β . Maximizing the product is not the same as maximizing $L(\beta)$, and the indicators carry information about β : the parameter spaces are not a product set.

41.3.4.2 B. Practice

Exercise. B1

Let (u_i, v_i) be a bivariate normal sample with u recorded for all n units and v for the first k , the mechanism depending on u but not on v . Reparameterizing by (μ_u, σ_u^2) and the regression parameters of v on u , derive the maximum likelihood estimates and the formula for $\hat{\mu}_v$ quoted above. Extend to three variables with a monotone pattern: u recorded n times, v the first k_2 , z the first $k_3 \leq k_2$.

Solution. Factor as $f(u)f(v | u)$; by Exercise (B1) an unrecorded v_i loses its conditional factor, so $L = \prod_{i \leq n} f(u_i) \prod_{i \leq k} f(v_i | u_i)$. The two parameter groups are variation independent, so each product is maximized alone: a normal sample of size n , and a normal linear model on k observations (Theorem 7.3.1). Since $\mu_v = \alpha + \beta\mu_u$ and $\hat{\alpha} = \bar{v}_{(k)} - \hat{\beta}\bar{u}_{(k)}$, and since a maximum likelihood estimate transforms with the parameterization, $\hat{\mu}_v$ is as stated. For three variables the same argument on $f(u)f(v | u)f(z | u, v)$ gives a univariate fit on all n , the regression of v on u over the first k_2 , and the regression of z on (u, v) over the first k_3 .

Exercise. B2

A linear model has a single covariate x that is sometimes missing, with $x \sim N(\mu, \tau^2)$ and $y | x \sim N(\beta_0 + \beta_1 x, \sigma^2)$. Write the observed-data likelihood contribution of a record with y recorded and x missing, and show that it is the density of a normal with mean $\beta_0 + \beta_1 \mu$ and variance $\beta_1^2 \tau^2 + \sigma^2$. Why does this contribution carry information about β_1 even though it contains no x ?

41.3.4.3 C. Going deeper

Exercise. C1

Show that Theorem 18.6.1 is the special case of [Theorem 41.3.1](#) in which the missing entries are responses, the mechanism depends only on the design, and the complete-data model is the normal linear model. Which part of Theorem 18.6.1 corresponds to [Eq. 41.3.2](#), and which to the warning that the degrees of freedom must be reduced?